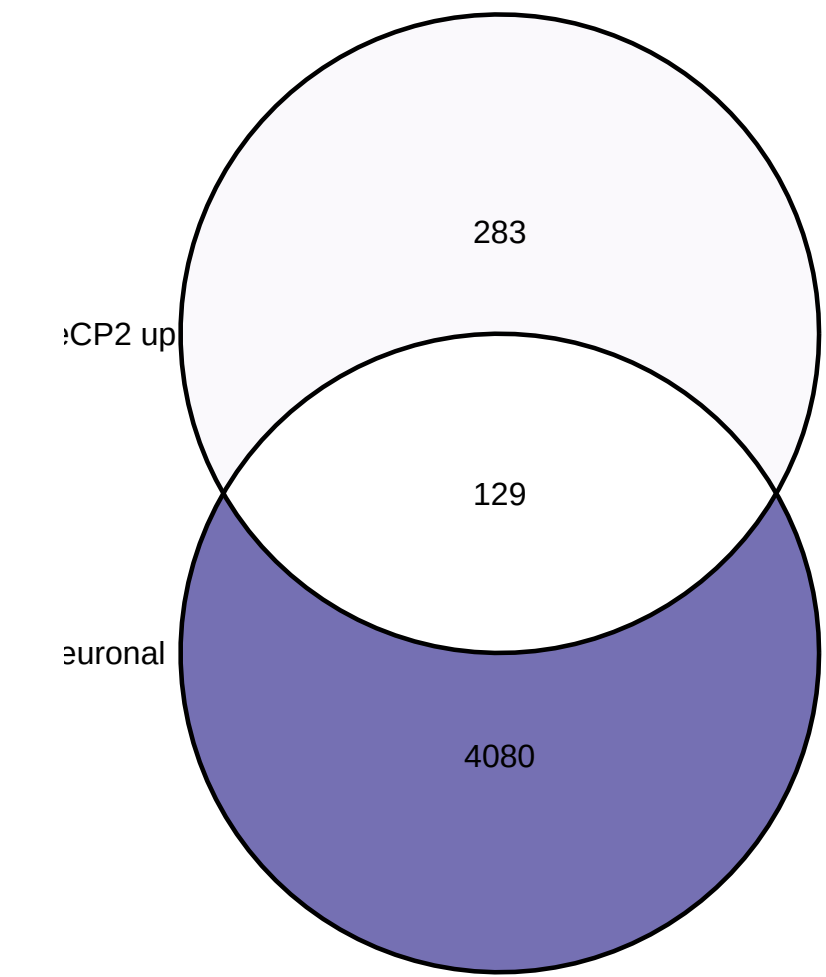


Neuronal genes and genes that gain MeCP2 binding

Universe: 21,072 genes tested for mCH

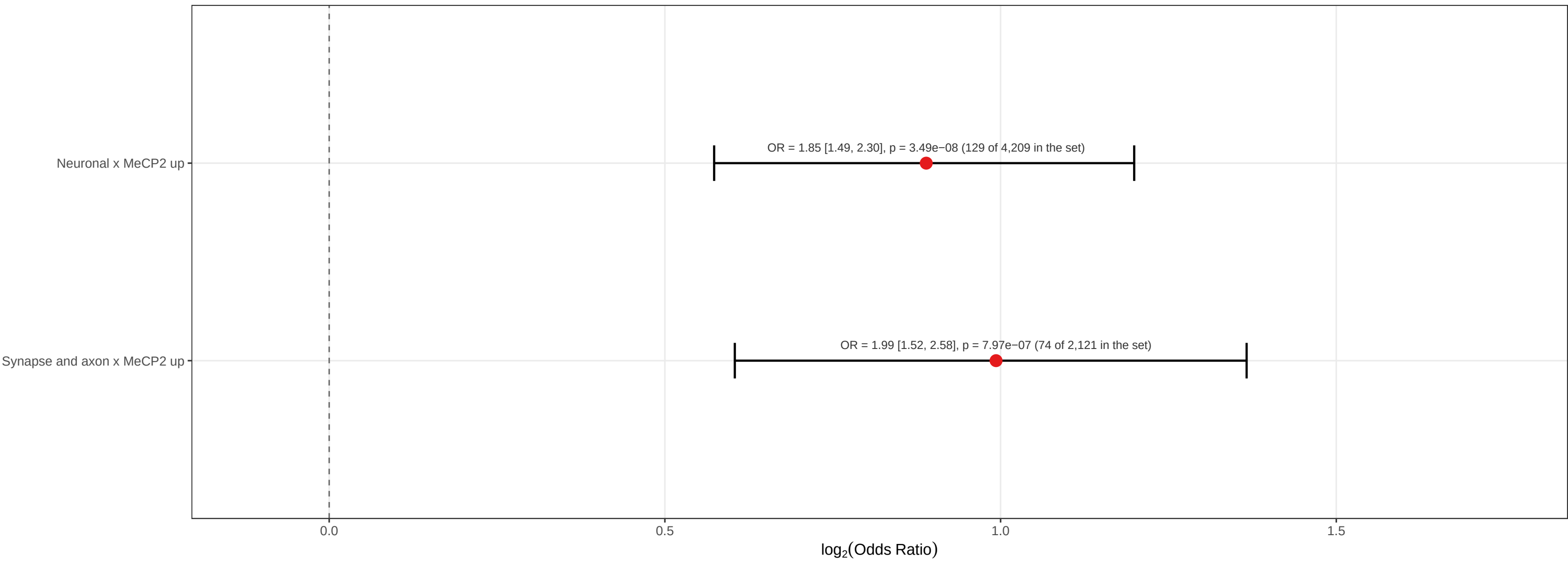


Neuronal only 4,080 | Both 129 | MeCP2 up only 283 | Neither 16,580
Fisher: OR = 1.85 [1.49, 2.30], p = 3.49e-08
3.1% of neuronal genes gain MeCP2, against 2.0% of the universe

Enrichment of MeCP2 gain in the neuronal gene sets

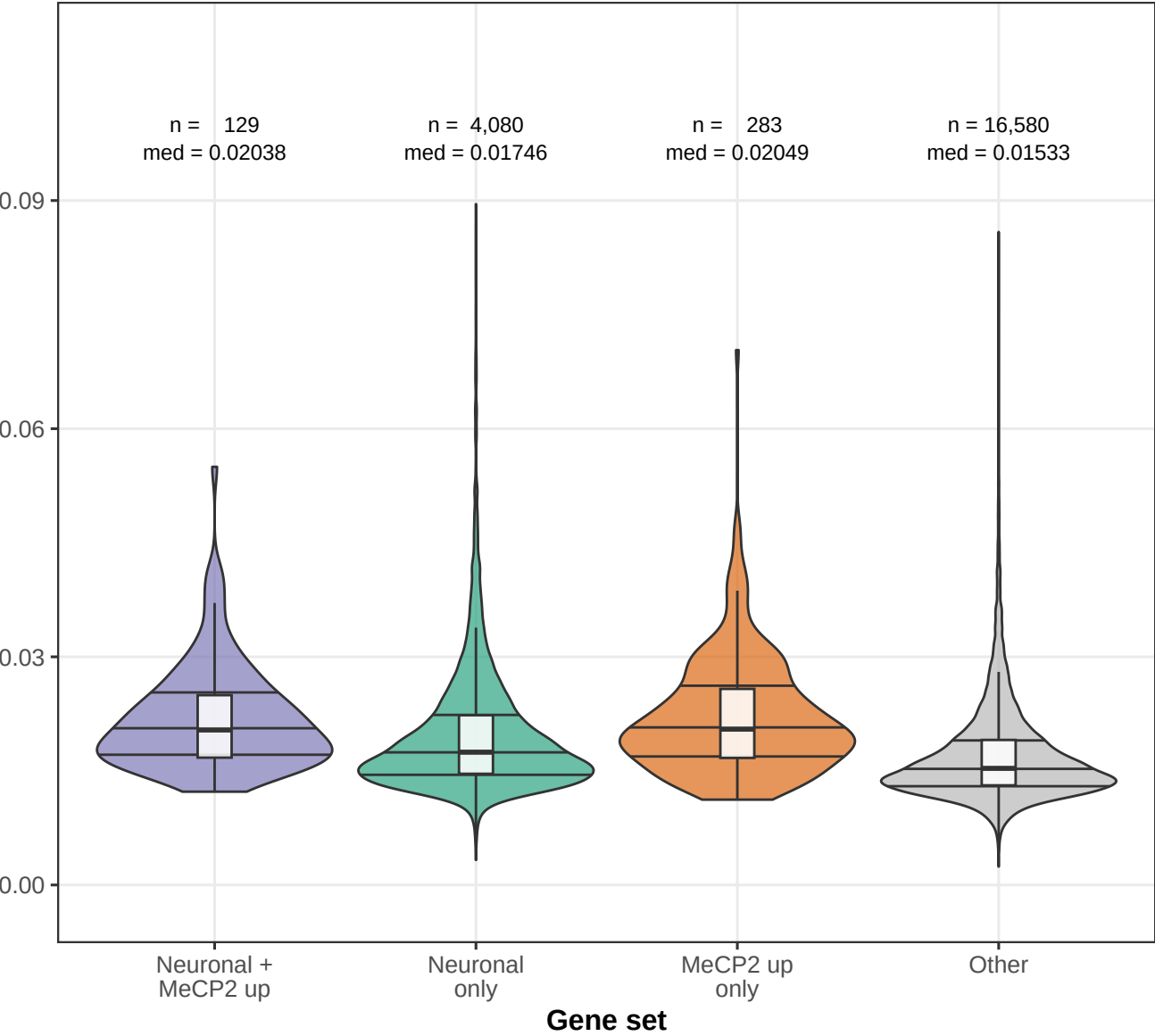
Universe: 21,072 genes tested for mCH | 412 gain MeCP2

Significant



Control mCH level by gene set

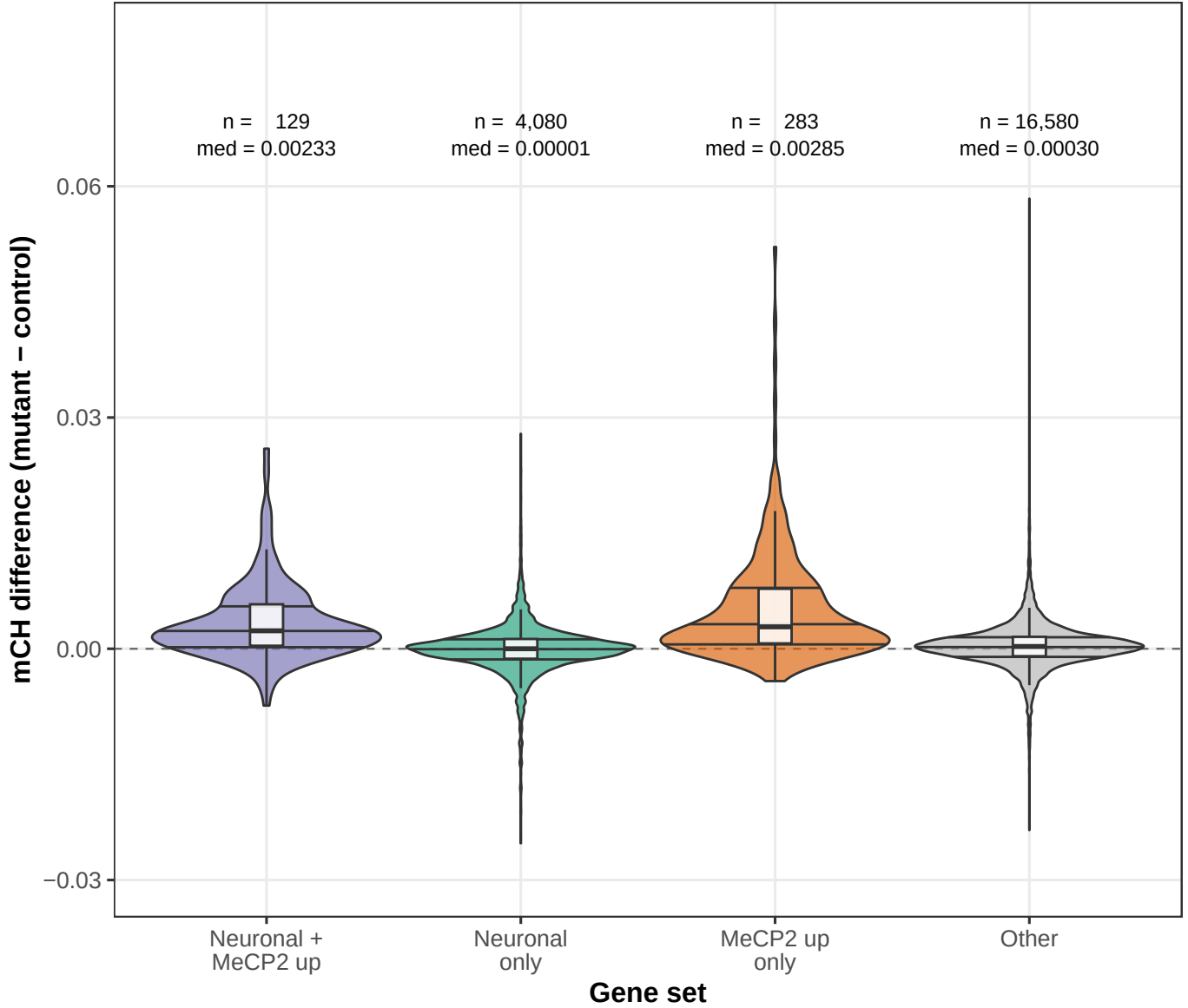
Gene-body non-CG methylation rate in the control condition



Wilcoxon Neuronal vs Other genes: p < 2.2e-16
Wilcoxon Synapse and axon vs Other genes: p < 2.2e-16
Wilcoxon Neuronal + MeCP2 up vs Neuronal only: p = 2.79e-07
Wilcoxon Neuronal + MeCP2 up vs MeCP2 up only: p = 1.00e+00

mCH change by gene set

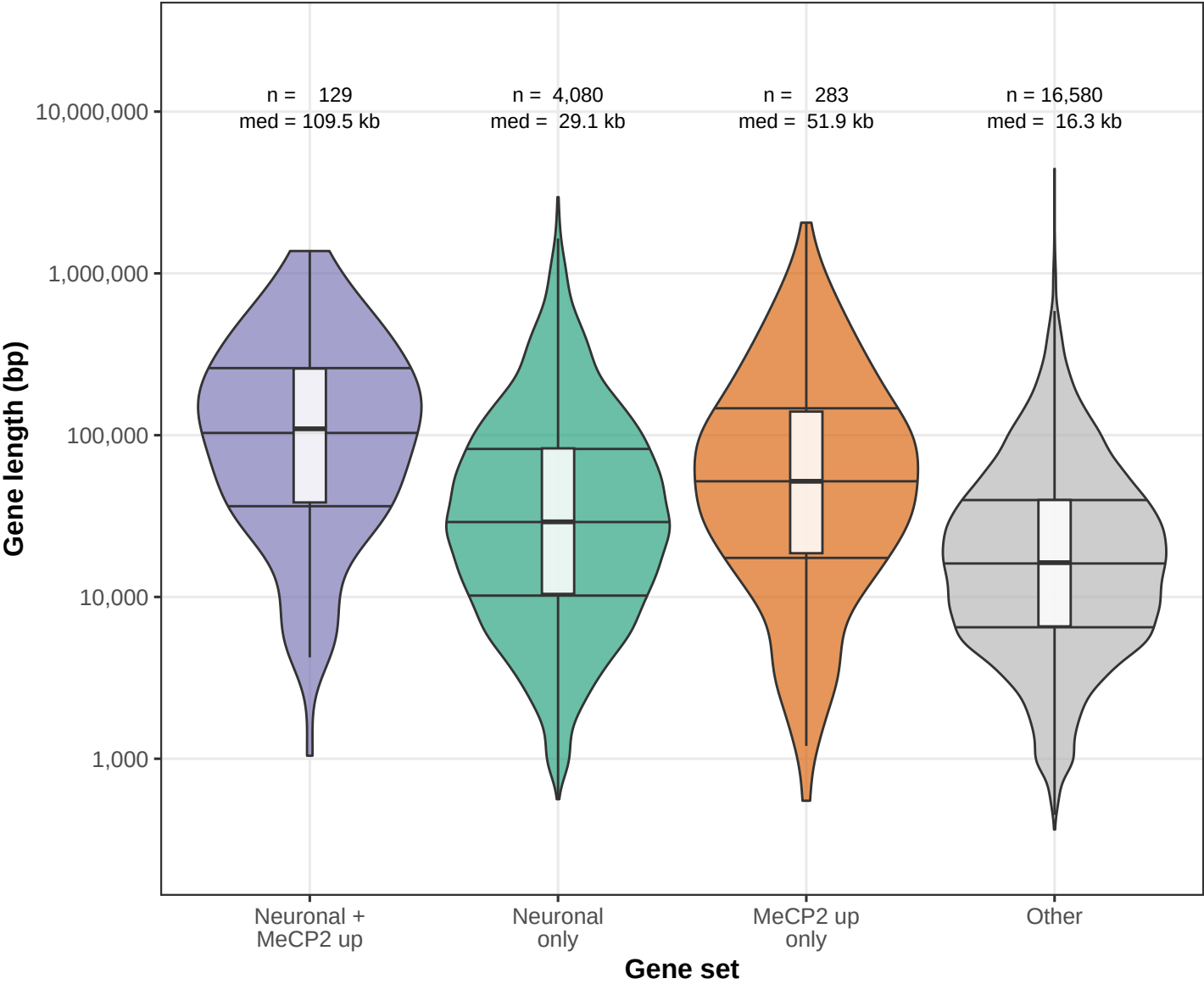
Mutant minus control gene-body mCH rate



Wilcoxon Neuronal vs Other genes: p < 2.2e-16
Wilcoxon Synapse and axon vs Other genes: p = 4.42e-11
Wilcoxon Neuronal + MeCP2 up vs Neuronal only: p < 2.2e-16
Wilcoxon Neuronal + MeCP2 up vs MeCP2 up only: p = 4.34e-02
Paired mutant vs control, Neuronal + MeCP2 up: median delta 0.00233, p = 7.85e-14
Paired mutant vs control, Neuronal only: median delta 0.00001, p = 4.45e-01
Paired mutant vs control, MeCP2 up only: median delta 0.00285, p < 2.2e-16
Paired mutant vs control, Other: median delta 0.00030, p < 2.2e-16

Gene length by gene set

Annotated gene body length, log10 axis



Gene length is associated with both mCH and MeCP2 binding, so it is shown for every set that the mCH panels compare.
Wilcoxon Neuronal vs Other genes: p < 2.2e-16
Wilcoxon Synapse and axon vs Other genes: p < 2.2e-16
Wilcoxon Neuronal + MeCP2 up vs Neuronal only: p < 2.2e-16
Wilcoxon Neuronal + MeCP2 up vs MeCP2 up only: p = 3.51e-05